

Gabriele Vianello

AI & Data Science Engineer

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Profile

Master's student in Computer Engineering (AI & Data Science). Passionate about Machine Learning, Zero-Trust architectures, and Data Engineering. Proven hands-on experience in designing ML pipelines, fine-tuning state-of-the-art models (LLM/VLM), and developing high-performance backend systems.

Technical Skills

AI & Data Science: PyTorch, TensorFlow, Hugging Face, Unsloth, Scikit-learn, Pandas, PyTorch Geometric, QLoRA
Programming: Python, Go, C#, TypeScript, C, SQL
Serving & DevOps: Docker, FastAPI, vLLM, ONNX Runtime, Ollama, MongoDB, PostgreSQL, DuckDB, GitHub Actions
Frontend: React, Vite, TailwindCSS, HTMX

Selected Projects

Graphagate | *AI Engineer & ML Researcher*

Temporal Graph Network for unsupervised anomaly detection on Zero-Trust access streams. Trained via negative sampling to score events in real-time, catching contextual, policy, and lateral-movement anomalies. Served as a FastAPI microservice. *Technologies: PyTorch, PyTorch Geometric, FastAPI, Docker*

ZTALeaks | *Security & AI Engineer*

Microservices reference implementation of a NIST SP 800-207 Zero Trust Architecture. Features strict PEP/PDP separation across isolated network segments, with access decisions combining OPA Rego policies and real-time AI risk scores. *Technologies: Go, OPA, Envoy, MongoDB, Docker, Snort*

synThor | *Data Engineer & ML Engineer*

Synthetic dataset generator for fine-tuning Vision-Language Models on document extraction. Produces perfectly aligned image and JSON ground-truth pairs across 4 document types and 11 languages, maintaining a fixed extraction schema. *Technologies: Python, PyMuPDF, ReportLab, Unsloth*

Pack-a-Punch | *AI Engineer*

Binary classifier distinguishing AI-generated from human-written Italian text, fine-tuned on bert-base-italian-xxl-cased. Inference pipeline optimized with ONNX Runtime and CUDA/Metal acceleration (~110 req/s). Fully containerized FastAPI service. *Technologies: Python, PyTorch, ONNX Runtime, FastAPI, Docker*

Charge-a-Sloth | *AI Engineer*

Optimized pipeline for fine-tuning multimodal vision-language models with Unsloth and QLoRA. Formats local parquet datasets into chat templates and runs 4-bit PEFT training on Blackwell-class GPUs. *Technologies: PyTorch, Unsloth, QLoRA, Hugging Face, CUDA*

Pack-a-Mail | *Backend Engineer*

High-throughput bulk email service written entirely in Go. Built with Clean Architecture (ports & adapters), shipping as a single static binary with an embedded HTMX web dashboard, CGO-free SQLite backend, and Argon2id hashing. *Technologies: Go, SQLite, HTMX, Clean Architecture*

Education

MSc in Computer Engineering (AI & Data Science)

Università Politecnica delle Marche

- Key courses: Deep Learning, Computer Vision, NLP, Data Mining, Network Science.

Current
Ancona, Italy

BSc in Engineering of Computing Systems

Politecnico di Milano

- Strong foundation in computer science, algorithms, and software architecture.

Completed
Cremona/Milano, Italy

High School Diploma (Computer Science)

Istituto Tecnico Industriale V. Volterra

Completed
Ancona, Italy

- Specialization in Computer Science and Telecommunications.